

Prevalence of HIV Drug Resistance and Its Determinants Among Unsuppressed Persons Living with HIV in Jamaica

Christine Ehikhamenor, Yonique Hanson, Karlidon Stewart, Kamelia Bryan, Andre Williams, Rahanah Khan-Francis, Tiffany Butterfield

Introduction

Jamaica has an estimated 28,000 persons living with HIV (PLHIV). The country has made steady progress in enrolling and retaining PLHIV in care on ART. While access to ART has increased over the last two decades, a growing global challenge has been the increasing prevalence of drug resistance. In Jamaica, few studies have examined HIVDR mutations, with none in the past decade. HIV drug resistance has grave implications for ART effectiveness, consequently impacting the health outcomes among the population.

Aim & Objective

The study aims to support clinical management and advance progress toward the third UNAIDS 90-90-90 target—viral suppression by describing the profile of unsuppressed PLHIV in Jamaica with HIVDR mutations.

Methods

A retrospective cross-sectional study of HIVDR samples of unsuppressed PLHIV in Jamaica was conducted between October 2024 to June 2025. A total of 212 plasma samples collected between 2018 and 2025 for HIV drug resistance (HIVDR) testing were reviewed. After clinical exclusions and laboratory quality control failures, 67 samples remained for analysis (Figure 1).

Samples collected were quantified using the Cobas® 6800 system, RNA extracted with the Invitrogen PureLink kit, and amplified with the AD4SEQ HIV-1 Solution v2. Libraries were sequenced on the Illumina MiSeq, and resistance interpretation was conducted with SmartVir software.

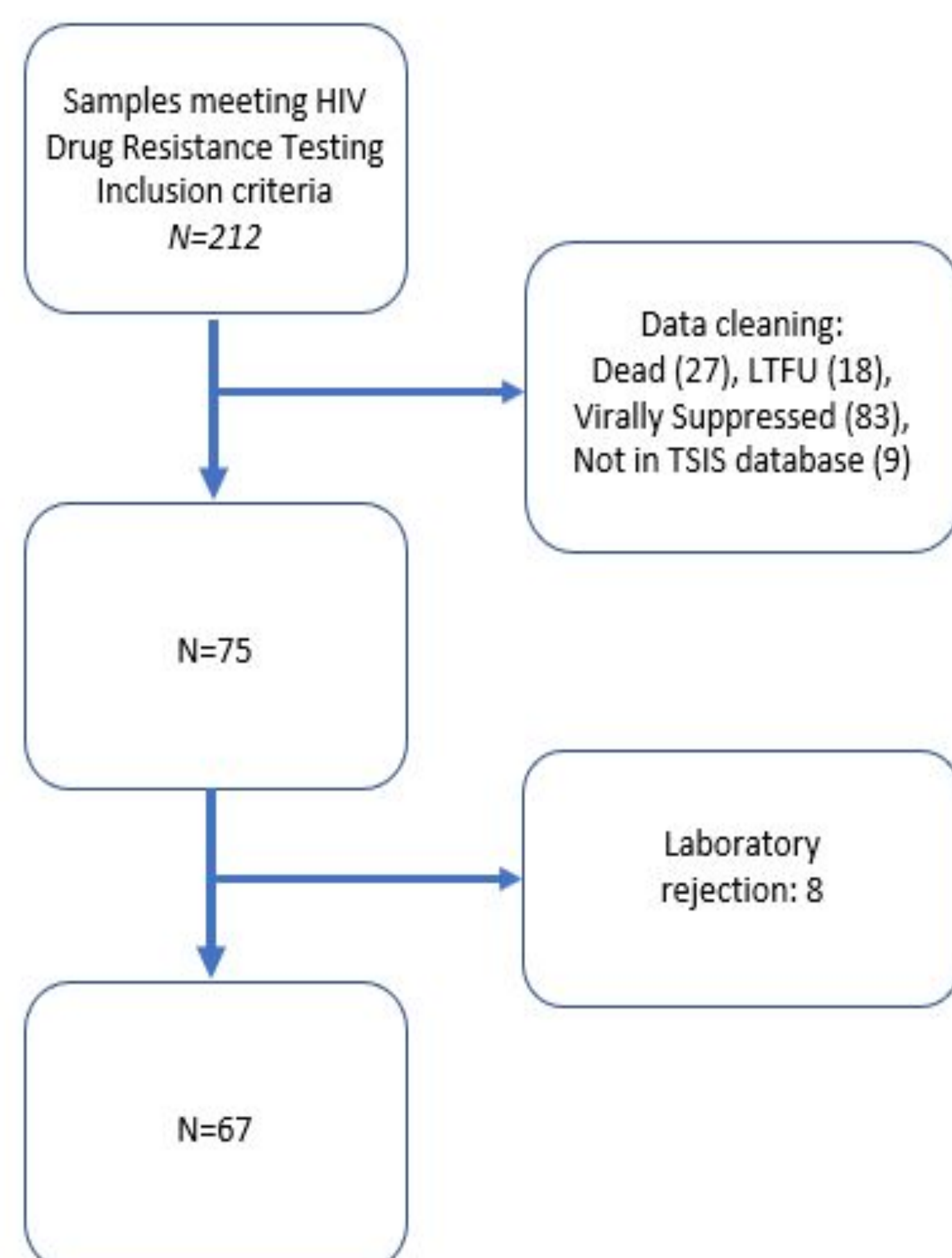


Figure 1. HIVDR Sample selection process

Statistical Analysis

We conducted statistical analysis using IBM SPSS version 20. Frequencies and proportions were used for categorical variables, and medians and means for continuous variables. Pearson's R test assessed associations between continuous variables, while the chi-square test examined associations between categorical variables.

Results

Baseline Characteristics and HIVDR

Of the 67 participants, 48% were male and 52% female, with a median age of 35 years. At ART initiation, 56% were classified at Stage III or IV (WHO staging). The median duration on ART at the time of HIVDR sample collection was 9 years (IQR: 7) (Table 1).

Results (cont'd)

HIVDR Mutations and resistance across drug classes

HIVDR mutations were present in all participants. Most had NNRTI mutations (99%), followed by NRTI mutations (76%), PI mutations (28%), and INSTI mutations (12%). HIVDR mutations spanned multiple drug classes. The majority of participants demonstrated resistance to multiple drug classes: 43% (n=29) had resistance to two drug classes, 31% (n=21) to three drug classes, 22% (n=15) to a single drug class, and 3% (n=2) exhibited resistance across all four drug classes (Table 1).

Factors associated with HIVDR

There were no significant associations found between the number of HIVDR mutations and the following variables: viral load ($r(67) = 0.009$, $p = 0.94$), and time on ART ($r(66) = 0.1$, $p = 0.424$). However, a significant inverse correlation was observed between lower CD4 counts at ART initiation and the number of HIVDR mutations ($r(64) = -0.292$, $p = 0.019$).

Table 1. Baseline participant characteristics and HIVDR

Characteristic	Frequency n (%) N=67
Sex	
Male	32 (48)
Female	35 (52)
Age Group	
<10 years	4 (6)
10-19 years	13 (19)
20-24 years	7 (10)
25-49 years	26 (39)
50+ years	17 (25)
Duration on ART (Years)	
<5 years	20 (30)
5-10 years	22 (33)
>10 years	24 (36)
Missing	1 (1)
VL Leading to the HIVDR test (copies/ml)	
1,000-9,999	13 (19)
10,000-99,999	34 (51)
100,000-999,999	16 (24)
1,000,000 +	4 (6)
WHO clinical stage (at ART Initiation)	
Stage I (> 500)	21 (31)
Stage II (350 – 499)	8 (12)
Stage III (200 – 349)	13 (19)
Stage IV (< 200)	25 (37)
HIVDR Mutations according to drug classes	
PIs	19 (28)
NRTIs	51 (76)
NNRTIs	66 (99)
INSTIs	8 (12)
Patterns of resistance across drug classes	
Single class	15 (22)
Dual classes	29 (43)
Triple class	21 (31)
Quadruple Class	2 (3)
Continuous Variable	
Characteristic	Median (IQR)
Age (years)	35 (33)
Duration on ART (years)	9 (7)
VL leading to HIVDR test (copies/mL)	34,000 (116,777)
Number of HIVDR mutations	5 (6)

Conclusions

Findings from the study indicate a high prevalence of drug resistant mutations, particularly among the Nucleotide Reverse Transcriptase Inhibitors. This resistance will likely result in increased ART treatment failure and limited regimen options. It is imperative that the national HIV response implement more robust approaches to routine resistance testing, viral load monitoring, and other PLHIV support systems that will mitigate HIV drug resistance and maintain ART efficacy.

Contact Information

Christine Ehikhamenor Christine.Ehikhamenor@moh.gov.jm
Yonique Hanson Yonique.hanson@gmail.com